

Breathing Difficulty: An In-Depth Analysis of Delhi's Air Quality Index (AQI)

Pollutant Dynamics, Diurnal Cycles, and the Geographic Drivers of Winter Pollution

Internship Project –**Intermediate Level**

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Internship Task: **An In-Depth Analysis of Delhi's Air Quality Index (AQI)**

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Executive Summary

This report presents an in-depth, data-driven analysis of Delhi's Air Quality Index (AQI) using an hourly, eight-pollutant monitoring dataset (CO, NO, NO₂, O₃, SO₂, PM_{2.5}, PM₁₀, and NH₃) spanning January 1–24, 2023. Applying the Central Pollution Control Board (CPCB) National AQI methodology, the analysis finds that every single day in the study window fell into the "Very Poor" or "Severe" category — 19 of 24 days (79%) were Severe (AQI > 400) and the remaining 5 were Very Poor (AQI 301–400). The mean daily AQI was 440, roughly nine times the CPCB "Good" threshold and far beyond the WHO 24-hour PM_{2.5} guideline, which every single day exceeded by a wide margin.

Fine particulate matter (PM_{2.5}) was the dominant pollutant on 21 of 24 days, with PM₁₀ dominant on the remaining 3 — confirming that particulates, not gaseous pollutants, are the primary driver of Delhi's winter air-quality crisis. Combustion-related pollutants (CO, NO, NO₂, PM_{2.5}, PM₁₀, NH₃) are very strongly inter-correlated ($r = 0.70\text{--}0.99$), pointing to shared sources — vehicular traffic, biomass/crop-residue burning, and other fossil-fuel combustion — while ozone (O₃) is negatively correlated with these primary pollutants, consistent with its photochemical, secondary formation and its titration by fresh NO emissions.

The diurnal (hour-of-day) profiles show a pronounced late-morning-to-evening pollution build-up (peaking around 15:00–17:00) and a shallower early-morning dip (07:00–09:00), a pattern this report ties to the interaction between traffic/combustion emission timing and the shrinking and expanding atmospheric mixing layer that characterizes Delhi's winter meteorology. Geographically, Delhi's landlocked position on the Indo-Gangetic Plain, encircled by higher terrain, combined with calm winter winds and frequent temperature inversions, traps pollutants close to the surface — a structural, geography-driven vulnerability that this report discusses in detail. The report closes with policy-relevant recommendations targeting the transport, construction, and biomass-burning sources implicated by the data.

1. Introduction: Delhi's Air Quality Challenge

Delhi, India's capital territory and one of the world's most populous urban agglomerations, experiences some of the highest ambient air pollution levels of any major city, especially during the winter months (November–January). This is not a coincidental or purely anthropogenic phenomenon — it is the product of a specific combination of geography, meteorology, and emission sources that converge each winter:

- **Geographic setting:** Delhi sits in the Indo-Gangetic Plain, a vast, relatively flat basin flanked by the Himalayan range to the north. This topography restricts the natural horizontal dispersion of pollutants and channels agricultural smoke from Punjab and Haryana toward the city.
- **Landlocked location:** unlike coastal cities, Delhi lacks the regular sea-breeze ventilation that helps flush pollutants; its climate is continental, with still, dry winter air.
- **Winter temperature inversions:** in winter, the ground cools faster than the air above it, creating a layer of warm air that sits atop cooler surface air (a temperature inversion). This suppresses vertical mixing and traps vehicular, industrial, and combustion emissions close to the ground overnight and in the early morning.
- **Low wind speeds:** winter is characterized by calm, low-wind conditions that reduce pollutant dispersal, unlike the monsoon and pre-monsoon seasons when wind and rain scavenge particulates.
- **Emission sources:** dense vehicular traffic, construction dust, industrial emissions, and — critically in the October–January window — large-scale crop-residue (stubble) burning in neighboring Punjab and Haryana, whose smoke is transported into Delhi's air-shed.

Given this context, the present report analyzes an hourly air-quality dataset for Delhi to characterize the magnitude, composition, and temporal dynamics of the pollution the city experiences, and to connect the statistical patterns found in the data back to these underlying geographic and meteorological drivers.

2. Dataset & Methodology

2.1 Data Description

The dataset comprises 561 hourly observations recorded from January 1, 2023 00:00 to January 24, 2023 08:00 (24 days), with no missing values. Eight pollutant concentrations are recorded for each hour:

- CO (Carbon Monoxide) — $\mu\text{g}/\text{m}^3$
- NO (Nitric Oxide) — $\mu\text{g}/\text{m}^3$
- NO₂ (Nitrogen Dioxide) — $\mu\text{g}/\text{m}^3$
- O₃ (Ozone) — $\mu\text{g}/\text{m}^3$
- SO₂ (Sulphur Dioxide) — $\mu\text{g}/\text{m}^3$
- PM_{2.5} (Fine Particulate Matter, $\leq 2.5 \mu\text{m}$) — $\mu\text{g}/\text{m}^3$
- PM₁₀ (Coarse Particulate Matter, $\leq 10 \mu\text{m}$) — $\mu\text{g}/\text{m}^3$
- NH₃ (Ammonia) — $\mu\text{g}/\text{m}^3$

2.2 AQI Calculation Method

Air Quality Index values were computed following the CPCB National AQI (2014) sub-index methodology. For each pollutant, a linear sub-index is calculated within the pollutant's concentration breakpoint band, and the overall AQI for a given period is the maximum of the individual pollutant sub-indices (the "dominant pollutant" is whichever pollutant produced that maximum). The breakpoints used are summarized below (condensed to four

of the eight pollutants for brevity; the analysis used the full CPCB table for all pollutants where breakpoints are published).

AQI Category (Range)	PM2.5 24h ($\mu\text{g}/\text{m}^3$)	PM10 24h ($\mu\text{g}/\text{m}^3$)	NO2 24h ($\mu\text{g}/\text{m}^3$)	CO 8h (mg/m^3)
Good (0–50)	0–30	0–50	0–40	0–1.0
Satisfactory (51–100)	31–60	51–100	41–80	1.1–2.0
Moderate (101–200)	61–90	101–250	81–180	2.1–10
Poor (201–300)	91–120	251–350	181–280	10.1–17
Very Poor (301–400)	121–250	351–430	281–400	17.1–34
Severe (401–500)	>250	>430	>400	>34

Table 1. Condensed CPCB AQI breakpoints used in sub-index calculation.

Daily AQI values in this report were computed from the mean pollutant concentration for each calendar day (approximating the CPCB's prescribed 24-hour / 8-hour averaging periods, given the hourly granularity of the source data). Hourly sub-indices were also computed for the diurnal and time-series analyses to preserve intra-day resolution.

2.3 Analytical Approach

The analysis combines descriptive statistics, CPCB sub-index computation, Pearson correlation analysis, and hour-of-day aggregation to answer four research questions:

#	Research Question	Method Used
RQ1	How severe is Delhi's air quality in January 2023, and how often are CPCB "Poor"/"Very Poor"/"Severe" thresholds breached?	CPCB sub-index AQI calculation; category classification
RQ2	What pollutant(s) dominate the AQI, and which pollutants move together (shared sources)?	Sub-index dominance count; Pearson correlation matrix
RQ3	How does air quality vary by hour of day (diurnal cycle), and what does this reveal about traffic, combustion, and atmospheric mixing?	Hourly grouping / diurnal profile plots
RQ4	How do geography and winter meteorology (temperature inversion, low wind, the Indo-Gangetic Plain, crop-residue burning) plausibly explain the observed patterns?	Literature-informed interpretation of the statistical patterns

3. Results

3.1 Overall Air Quality Status (RQ1)

Across all 24 days in the sample, not a single day achieved a rating better than "Very Poor." The daily AQI averaged 440 (CPCB "Severe" territory begins at 401), ranging from a minimum of 306 (Jan 15, still "Very Poor") to a maximum of 635 (Jan 2, deep into "Severe"). Nineteen of twenty-four days (79%) were classified Severe and the remaining five (21%) Very Poor — meaning the entire observation window fell in the two worst of CPCB's six AQI categories.

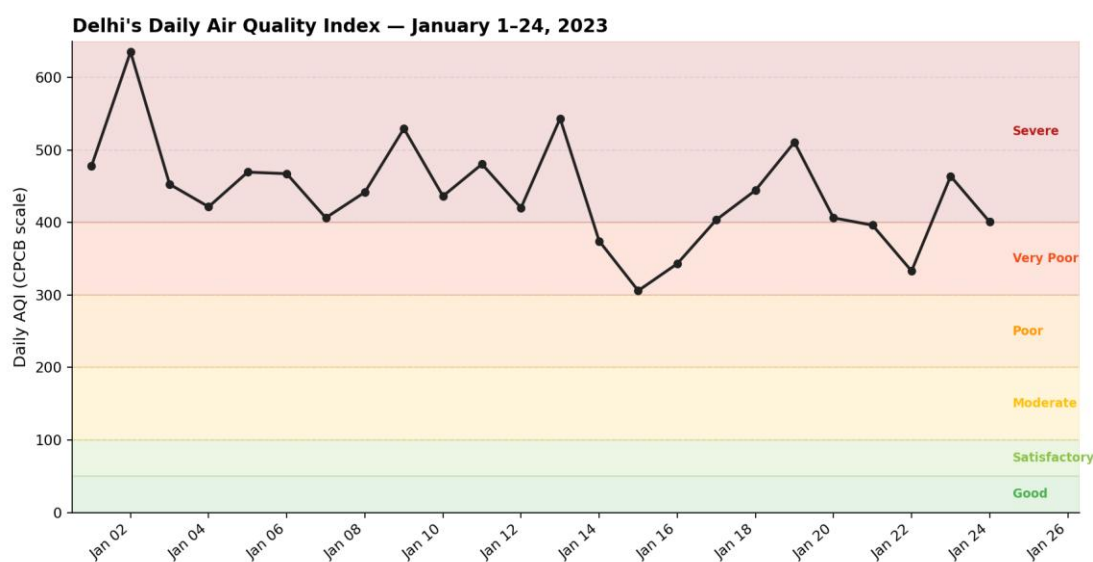


Figure 1. Daily AQI trend, January 1–24, 2023, with CPCB severity bands. Every day falls in the Very Poor or Severe band.

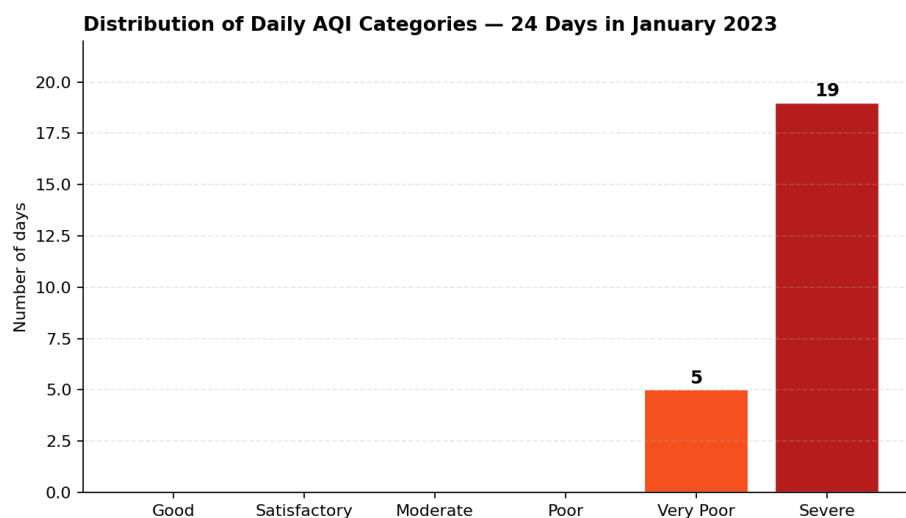


Figure 2. Distribution of daily AQI categories over the 24-day window.

Mean daily PM_{2.5} across the period was 355.5 µg/m³ and mean PM₁₀ was 418.0 µg/m³ — both roughly 6–12 times the CPCB 24-hour "Satisfactory" ceilings (60 and 100 µg/m³ respectively), and more than 20 times the WHO's much stricter 24-hour PM_{2.5} guideline of 15 µg/m³. Every one of the 24 days breached both the Indian NAAQS and the WHO PM_{2.5} guideline.

3.2 Pollutant Dominance and Correlation Structure (RQ2)

PM_{2.5} was the dominant (AQI-determining) pollutant on 21 of 24 days (87.5%), with PM₁₀ dominant on the remaining 3 days — no gaseous pollutant (NO₂, SO₂, CO, O₃, NH₃) ever drove the daily AQI in this dataset. This squarely implicates particulate matter, rather than any single gaseous pollutant, as the central public-health concern in Delhi's winter air.

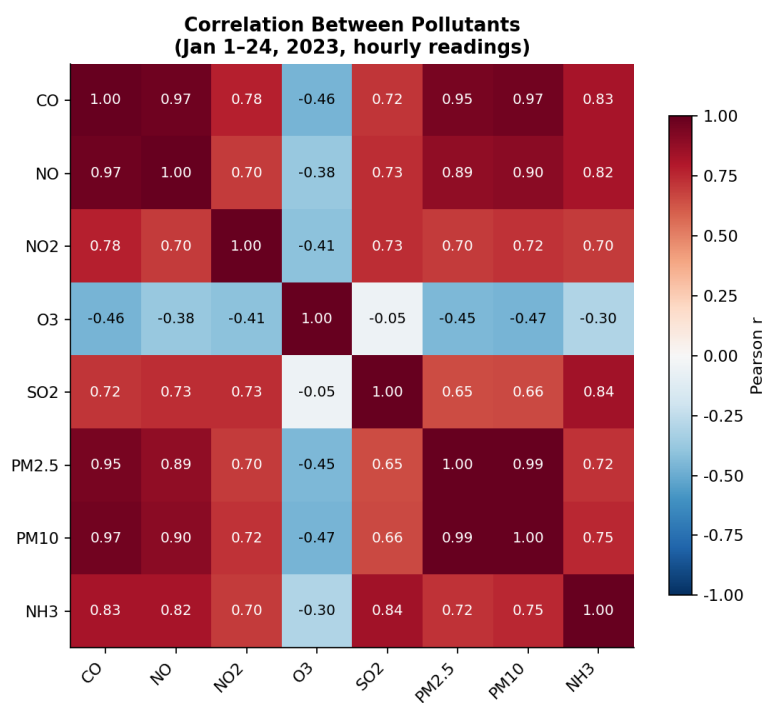


Figure 3. Pearson correlation matrix across all eight measured pollutants (hourly data).

The correlation matrix reveals two distinct pollutant groupings. First, a tightly-coupled combustion cluster — CO, NO, NO₂, PM_{2.5}, PM₁₀, and NH₃ — shows strong positive correlations ($r = 0.70$ to 0.99) with one another. CO and PM₁₀ are almost perfectly correlated ($r = 0.97$), as are PM_{2.5} and PM₁₀ ($r = 0.99$), indicating these pollutants largely share common combustion and re-suspension sources (vehicle exhaust, biomass burning, and dust). Second, ozone (O₃) stands apart, correlating negatively with every primary pollutant ($r = -0.30$ to -0.47). This is consistent with the well-established atmospheric chemistry of ozone: it is a secondary pollutant formed photochemically from NO_x and VOCs under sunlight, and it is rapidly consumed (titrated) by fresh NO emissions — so where combustion pollutants are highest (early morning, evening), O₃ tends to be at its lowest, and vice versa.

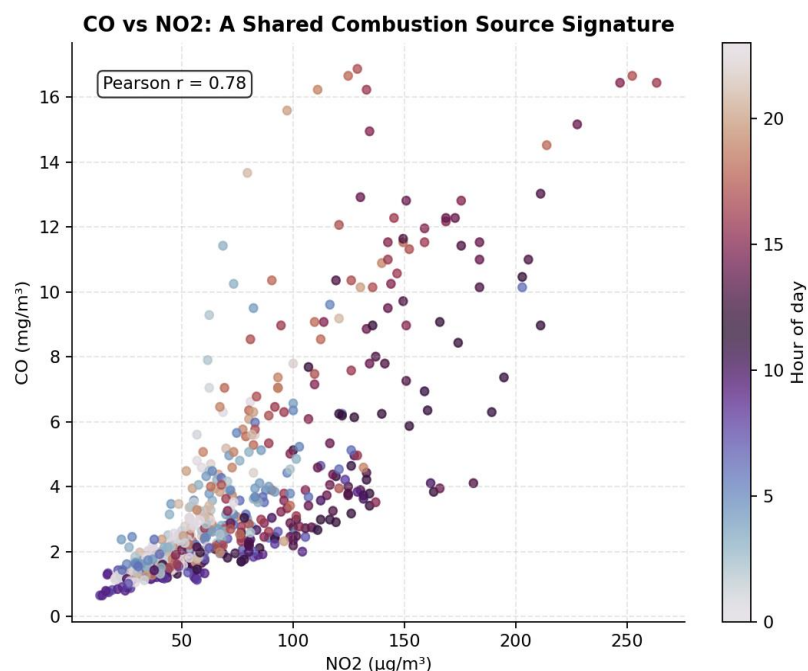


Figure 4. CO vs NO₂ scatter, colored by hour of day. The strong, roughly linear relationship ($r = 0.78$) is a signature of a shared combustion source (traffic, and other fossil-fuel burning).

3.3 Diurnal (Hour-of-Day) Patterns (RQ3)

Aggregating all 24 days by hour of day reveals a consistent and pronounced daily cycle across the combustion-linked pollutants. PM_{2.5} and PM₁₀ are lowest in the mid-morning window (07:00–09:00, PM_{2.5} $\approx$ 195–216 $\mu\text{g}/\text{m}^3$) and build steadily to a peak in the afternoon-to-evening window (15:00–17:00, PM_{2.5} $\approx$ 530–540 $\mu\text{g}/\text{m}^3$) — roughly a 2.7x swing between the daily low and high. CO and NO₂ follow a closely related but slightly different rhythm: both are elevated overnight (00:00–06:00), dip sharply around 07:00–09:00, and then climb to a pronounced midday-to-afternoon peak (CO peaks near 15:00, NO₂ peaks near 13:00) before declining through the evening.

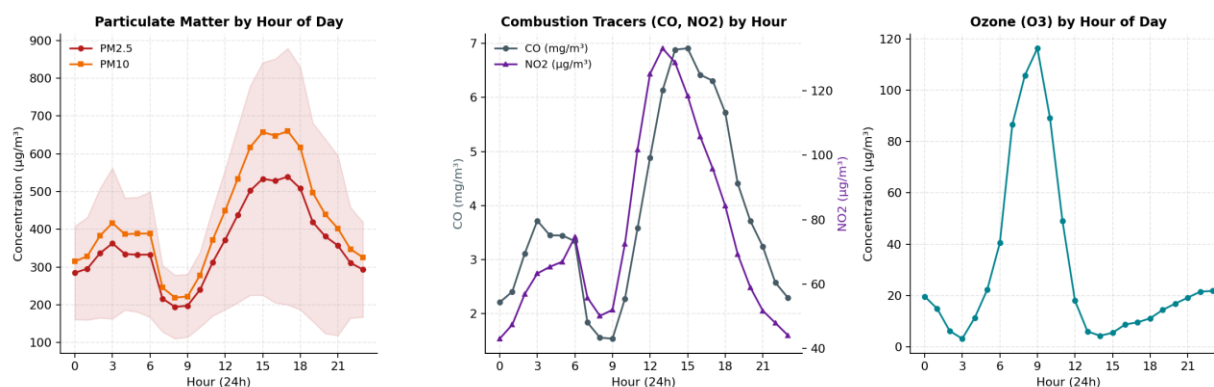


Figure 5. Average diurnal (hour-of-day) profiles for particulate matter, combustion tracers (CO, NO₂), and ozone, averaged across all 24 days.

Ozone shows the inverse rhythm expected of a photochemically-formed pollutant: it is near zero overnight, begins rising after sunrise, peaks around mid-morning (09:00–10:00, $\approx$ 89–116 $\mu\text{g}/\text{m}^3$) as sunlight drives

photochemical production, and then collapses again by early afternoon as it is scavenged by rising NO_x from midday traffic and combustion, staying low through the night.

The observed shape — a late-morning dip and a broad afternoon/evening peak, rather than the classic sharp "double rush-hour hump" often reported for ground monitoring stations — is discussed further in Section 4, since it is best explained jointly by emission timing and the daily cycle of the atmospheric mixing layer rather than by traffic volume alone.

3.4 Episodes and Day-to-Day Variability

While the entire period was severely polluted, day-to-day AQI still varied meaningfully (from 306 to 635), showing that even within a persistently polluted regime, short-term meteorological variability (wind speed, humidity, temperature inversion strength) modulates severity. The worst single day, January 2 (AQI 635, PM₁₀-dominant), and a cluster of severe days January 9–13, likely correspond to spells of calm winds and strong inversions — conditions known to produce Delhi's periodic "pollution episodes" and smog events. The comparatively "better" days (Jan 14–16 and Jan 21–22, still Very Poor) likely coincide with somewhat stronger wind or partial dispersal, though confirming this precisely would require pairing the pollutant data with meteorological records (wind speed/direction, boundary-layer height), which were outside the scope of this dataset.

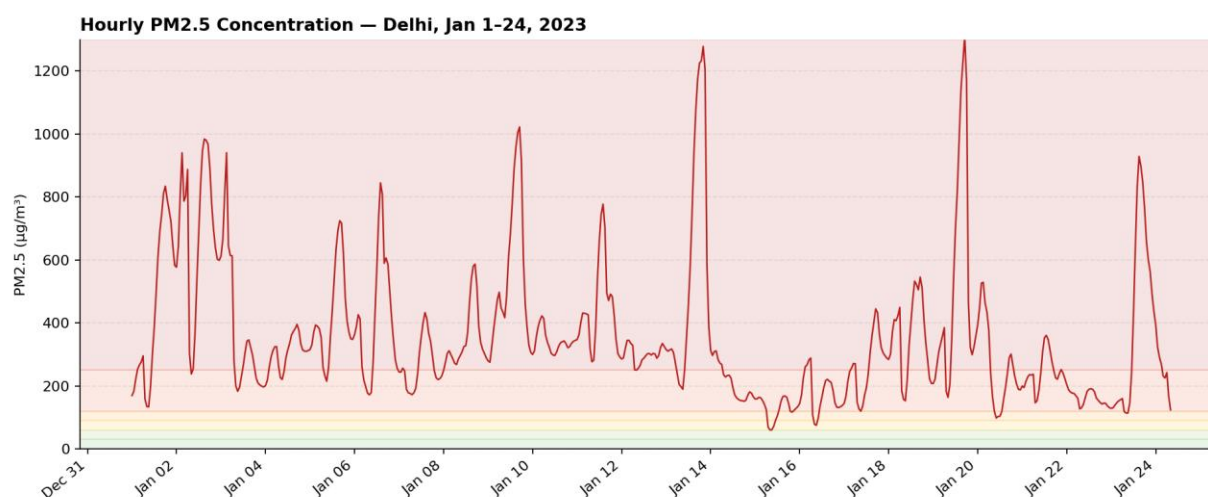


Figure 6. Full-resolution hourly PM_{2.5} series showing both the diurnal cycle and multi-day episodes of especially severe pollution.

3.5 Daily AQI Summary Table

Date	Mean PM2.5 ($\mu\text{g}/\text{m}^3$)	AQI	Category	Dominant Pollutant
2023-01-01	443.9	478	Severe	PM2.5
2023-01-02	698.1	635	Severe	PM10
2023-01-03	381.8	453	Severe	PM2.5
2023-01-04	304.0	422	Severe	PM2.5
2023-01-05	423.6	469	Severe	PM2.5
2023-01-06	418.1	467	Severe	PM2.5
2023-01-07	265.9	406	Severe	PM2.5
2023-01-08	354.2	442	Severe	PM2.5
2023-01-09	557.8	530	Severe	PM10
2023-01-10	340.8	436	Severe	PM2.5
2023-01-11	451.0	480	Severe	PM2.5
2023-01-12	300.4	420	Severe	PM2.5
2023-01-13	590.4	543	Severe	PM10
2023-01-14	216.7	374	Very Poor	PM2.5
2023-01-15	127.7	306	Very Poor	PM2.5
2023-01-16	176.0	343	Very Poor	PM2.5
2023-01-17	259.3	404	Severe	PM2.5
2023-01-18	360.4	444	Severe	PM2.5
2023-01-19	526.5	511	Severe	PM2.5
2023-01-20	265.8	406	Severe	PM2.5
2023-01-21	245.0	396	Very Poor	PM2.5
2023-01-22	162.9	333	Very Poor	PM2.5
2023-01-23	409.5	464	Severe	PM2.5
2023-01-24	251.8	401	Severe	PM2.5

Table 2. Computed daily AQI, category, and dominant pollutant for each day in the study period.

4. Discussion: Geographic and Meteorological Drivers (RQ4)

4.1 Why Winter, and Why Delhi

The statistical signature found in this dataset — extremely elevated particulate matter, a tight combustion-pollutant correlation cluster, and an inverse ozone relationship — is fully consistent with the well-documented winter smog phenomenon that afflicts Delhi and the broader Indo-Gangetic Plain each year. Three geographic/meteorological mechanisms plausibly explain the patterns observed:

- **Temperature inversion and shallow mixing layer:** In winter, radiative cooling at the surface after sunset creates a stable air layer that suppresses vertical mixing. Pollutants emitted by traffic, industry, and heating/biomass burning become trapped close to the ground rather than dispersing upward, which is consistent with the elevated overnight and early-morning CO and NO₂ levels seen in the data before the inversion begins breaking down mid-morning.
- **Boundary-layer expansion and re-trapping cycle:** As the sun heats the surface through the morning, the mixing layer deepens and temporarily dilutes near-surface pollutant concentrations — plausibly contributing to the 07:00–09:00 dip observed in PM_{2.5}/PM₁₀/CO. As the afternoon progresses and emissions continue (traffic, cooking, construction, evening biomass burning) while the mixing layer begins to shallow again ahead of sunset, pollutants re-accumulate — consistent with the pronounced 15:00–18:00 peak found in this dataset.
- **Topography and regional transport:** Delhi's position on the flat Indo-Gangetic Plain, without significant natural barriers to funnel pollution away, combined with calm winter wind patterns, limits the horizontal transport of pollutants out of the region. During the October–January window, this is compounded by smoke from large-scale agricultural residue burning in the neighboring states of Punjab and Haryana, which is carried into the Delhi air-shed by prevailing northwesterly winds — a well-established contributor to the city's worst winter pollution spikes.

It should be noted that the precise diurnal shape found here (a single broad afternoon peak, rather than distinct morning- and evening-rush humps) can also reflect characteristics of the specific data source and its spatial/temporal resolution; ground-based continuous monitoring stations in Delhi have at times reported a more bimodal (two-peak) rush-hour pattern. The mechanisms above remain the most plausible explanation, but pairing this data with local meteorological records (wind speed, boundary-layer height, humidity) would allow the mixing-layer explanation to be tested directly rather than inferred.

4.2 Public Health Implications

Every day in the sample exceeded CPCB's "Very Poor" threshold, a category associated with respiratory illness on prolonged exposure, and the majority of days reached "Severe," CPCB's designation for conditions that can affect healthy people and seriously impact those with existing respiratory or cardiovascular disease. Given that PM_{2.5} — the pollutant most strongly linked in the epidemiological literature to cardiovascular and respiratory mortality — was the dominant pollutant on 87.5% of days, sustained exposure of this kind over a full winter season represents a substantial public-health burden for Delhi's population, particularly for children, the elderly, outdoor workers, and those with pre-existing conditions.

5. Limitations

- **Temporal coverage:** the dataset spans only 24 days within a single winter month (January 2023). This precludes any direct analysis of seasonal variation (e.g., comparison against Delhi's cleaner monsoon or post-monsoon periods) — a research question named in the assignment brief that could not be addressed

with this dataset alone; all seasonal context here is drawn from established literature rather than the data itself.

- No meteorological variables: wind speed/direction, temperature, humidity, and boundary-layer height were not included, so the meteorological explanations offered in Section 4 are literature-informed interpretations of the observed statistical patterns rather than directly tested relationships.
- Single (city-level) data source: the dataset does not distinguish between monitoring stations, so intra-city / neighborhood-level spatial variation in Delhi (which can be substantial) is not captured.
- AQI averaging approximation: CPCB's official methodology uses strict 24-hour (PM2.5, PM10, NO2, SO2, NH3) and 8-hour (CO, O3) rolling averages with minimum data-completeness rules; this analysis approximates that using full calendar-day and full-dataset means, which is a reasonable but not identical approximation.

6. Conclusion & Recommendations

This analysis confirms, using granular hourly data, that Delhi experienced a sustained, severe air pollution crisis throughout the studied period in January 2023, with every day rated Very Poor or Severe and fine particulate matter (PM2.5) as the overwhelming driver. The tight correlation among combustion-linked pollutants points to overlapping sources — chiefly vehicular traffic and biomass/solid-fuel combustion — while the diurnal cycle and the geographic/meteorological context together explain why these emissions accumulate to such damaging levels specifically in winter. Based on these findings, the following targeted interventions are suggested as consistent with the pollutant and temporal patterns identified:

- Prioritize particulate-matter control (PM2.5/PM10) over any single gaseous pollutant, since PM2.5 alone determined the AQI on 87.5% of days — e.g., stricter dust-control at construction sites, mechanized road sweeping, and continued restriction of high-emission vehicles.
- Target emission-reduction measures (traffic restrictions, industrial curbs) toward the afternoon build-up window (roughly 12:00–18:00) and the overnight/early-morning inversion window (00:00–07:00), when the data show pollutant accumulation is worst.
- Continue and strengthen regional coordination on crop-residue burning with neighboring states, given the well-documented contribution of transported agricultural smoke to Delhi's winter particulate load.
- Expand monitoring to include meteorological co-variates (wind, boundary-layer height) at the same stations, enabling future analyses to directly attribute day-to-day AQI swings to specific dispersion conditions rather than inferring them.
- Extend data collection across a full year, enabling a genuine seasonal comparison (winter vs. monsoon vs. summer) to quantify how much of Delhi's pollution burden is seasonal/meteorological versus a year-round structural issue.